

$$\begin{aligned}
& \frac{2\lambda}{2\pi^2} \left(\frac{1}{4} m_1 \tilde{\mu} g_1^4 \text{LF}_{2,2,0}[m_1, \tilde{\mu}] - \frac{3}{2} m_1 \tilde{\mu} g_1^4 \text{LF}_{3,2,-1}[m_1, \tilde{\mu}] + m_1 \tilde{\mu} g_1^4 \text{LF}_{3,3,-2}[m_1, \tilde{\mu}] + \right. \\
& m_1 \tilde{\mu} g_1^4 \text{LF}_{4,2,-2}[m_1, \tilde{\mu}] - \frac{1}{4} m_2 \tilde{\mu} g_2^4 \text{LF}_{2,2,0}[m_2, \tilde{\mu}] - 4 m_2 \tilde{\mu} g_2^4 \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] + \\
& 3 m_2 \tilde{\mu} g_2^4 \text{LF}_{3,3,-2}[m_2, \tilde{\mu}] + 3 m_2 \tilde{\mu} g_2^4 \text{LF}_{4,2,-2}[m_2, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,1,0}[m_d^{\text{r}}, m_q^{\text{p}}] - \\
& \frac{1}{4} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,2,-1}[m_d^{\text{r}}, m_q^{\text{p}}] - \frac{1}{2} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{4,1,-1}[m_d^{\text{r}}, m_q^{\text{p}}] + \\
& \frac{1}{2} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,1,0}[m_e^{\text{r}}, m_l^{\text{p}}] - \frac{1}{4} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,2,-1}[m_e^{\text{r}}, m_l^{\text{p}}] - \\
& \frac{1}{2} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{4,1,-1}[m_e^{\text{r}}, m_l^{\text{p}}] - \frac{1}{8} \tilde{\mu} y_e^{\text{pr}} \overline{a_e^{\text{pr}}} (g_1^2 + 5 g_2^2) \text{LF}_{4,1,-1}[m_l^{\text{p}}, m_e^{\text{r}}] + \\
& \frac{1}{2} \tilde{\mu} g_2^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{5,1,-2}[m_l^{\text{p}}, m_e^{\text{r}}] + \frac{1}{8} \tilde{\mu} y_d^{\text{pr}} \overline{a_d^{\text{pr}}} (g_1^2 - 15 g_2^2) \text{LF}_{4,1,-1}[m_q^{\text{p}}, m_d^{\text{r}}] + \\
& \frac{3}{2} \tilde{\mu} g_2^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{5,1,-2}[m_q^{\text{p}}, m_d^{\text{r}}] - \frac{3}{4} \tilde{\mu} g_2^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,0}[m_q^{\text{p}}, m_u^{\text{r}}] + \\
& \frac{1}{4} \tilde{\mu} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 + 6 g_2^2) \text{LF}_{3,1,0}[m_q^{\text{p}}, m_u^{\text{r}}] - \frac{1}{8} \tilde{\mu} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 + 3 g_2^2) \text{LF}_{3,2,-1}[m_q^{\text{p}}, m_u^{\text{r}}] - \\
& \frac{1}{4} \tilde{\mu} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 + 6 g_2^2) \text{LF}_{4,1,-1}[m_q^{\text{p}}, m_u^{\text{r}}] + \frac{3}{4} \tilde{\mu} g_2^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,1,0}[m_u^{\text{r}}, m_q^{\text{p}}] + \\
& \frac{3}{4} \tilde{\mu} g_2^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,2,-1}[m_u^{\text{r}}, m_q^{\text{p}}] - \frac{1}{4} \tilde{\mu} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (2 g_1^2 + 9 g_2^2) \text{LF}_{4,1,-1}[m_u^{\text{r}}, m_q^{\text{p}}] + \\
& \frac{3}{2} \tilde{\mu} g_2^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{5,1,-2}[m_u^{\text{r}}, m_q^{\text{p}}] - \frac{3}{8} m_1 \tilde{\mu} g_1^4 \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] - \frac{3}{8} m_1 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{4,1,-1}[\tilde{\mu}, m_1] + \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^4 \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{2} m_1 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{5,1,-2}[\tilde{\mu}, m_1] + m_2 \tilde{\mu} g_2^4 \text{LF}_{3,1,0}[\tilde{\mu}, m_2] + \\
& \frac{5}{8} m_2 \tilde{\mu} g_2^4 \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] - \frac{21}{8} m_2 \tilde{\mu} g_2^4 \text{LF}_{4,1,-1}[\tilde{\mu}, m_2] - \frac{3}{4} m_2 \tilde{\mu} g_2^4 \text{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \\
& \frac{3}{2} m_2 \tilde{\mu} g_2^4 \text{LF}_{5,1,-2}[\tilde{\mu}, m_2] + \frac{1}{2} \tilde{\mu} g_1^2 g_2^2 (7 m_1 + 5 m_2) \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
& 2 \tilde{\mu} g_1^2 g_2^2 (m_1 + m_2) \text{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] - \frac{3}{2} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \text{LF}_{2,1,1,0}[m_d^{\text{r}}, m_d^{\text{t}}, m_q^{\text{p}}] + \\
& \frac{3}{2} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,1,1,-1}[m_d^{\text{r}}, m_d^{\text{t}}, m_q^{\text{p}}] + \\
& \frac{3}{4} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \text{LF}_{2,2,1,-1}[m_d^{\text{r}}, m_q^{\text{p}}, m_d^{\text{t}}] - \frac{3}{2} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \\
& \text{LF}_{2,1,1,0}[m_d^{\text{t}}, m_d^{\text{r}}, m_q^{\text{p}}] + \frac{3}{2} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,1,1,-1}[m_d^{\text{t}}, m_d^{\text{r}}, m_q^{\text{p}}] + \\
& \frac{3}{4} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \text{LF}_{2,2,1,-1}[m_d^{\text{t}}, m_q^{\text{p}}, m_d^{\text{r}}] - \frac{1}{2} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \\
& \text{LF}_{2,1,1,0}[m_e^{\text{r}}, m_e^{\text{t}}, m_l^{\text{p}}] + \frac{1}{2} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,1,1,-1}[m_e^{\text{r}}, m_e^{\text{t}}, m_l^{\text{p}}] + \\
& \frac{1}{4} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \text{LF}_{2,2,1,-1}[m_e^{\text{r}}, m_l^{\text{p}}, m_e^{\text{t}}] - \frac{1}{2} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \\
& \text{LF}_{2,1,1,0}[m_e^{\text{t}}, m_e^{\text{r}}, m_l^{\text{p}}] + \frac{1}{2} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,1,1,-1}[m_e^{\text{t}}, m_e^{\text{r}}, m_l^{\text{p}}] + \\
& \frac{1}{4} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \text{LF}_{2,2,1,-1}[m_e^{\text{t}}, m_l^{\text{p}}, m_e^{\text{r}}] + \frac{1}{4} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \\
& \text{LF}_{2,2,1,-1}[m_l^{\text{p}}, m_l^{\text{s}}, m_e^{\text{r}}] + \frac{3}{4} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \text{LF}_{2,2,1,-1}[m_q^{\text{p}}, m_q^{\text{s}}, m_d^{\text{r}}] - \\
& \frac{9}{4} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,1,1,0}[m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}] + \frac{9}{4} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \\
& \text{LF}_{3,1,1,-1}[m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}] + \frac{9}{8} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,1,-1}[m_q^{\text{p}}, m_u^{\text{r}}, m_q^{\text{s}}] - \\
& \frac{9}{4} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,1,1,0}[m_q^{\text{s}}, m_q^{\text{p}}, m_u^{\text{r}}] + \frac{9}{4} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \\
& \text{LF}_{3,1,1,-1}[m_q^{\text{s}}, m_q^{\text{p}}, m_u^{\text{r}}] + \frac{9}{8} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,1,-1}[m_q^{\text{s}}, m_u^{\text{r}}, m_q^{\text{p}}] - \\
& m_1 \tilde{\mu} g_1^2 g_2^2 \text{$$